

ContextOS Governed Response

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ContextOS Envelope:

request_id : f5dab15b-52d2-44a0-a6e4-5c25927a28b1
source_app : ContextOS
target_app : LineAlertDemo
user_intent : Label Alignment Off
operation_scope : demo
evidence : ContextOS matched intent 'Label Alignment Off' to registered application 'LineAlertDemo'
assumptions : Registered application mapping is safe to invoke locally
decision : Proceed
confidence : Medium-high
tradeoff_summary : Registered intent proceeds through the selected application; policy remains limited to explicit registry mappings

Application Output:

Machine Context
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Machine:

Bottle Labeling Machine

Purpose:

Apply printed labels to bottles moving on a conveyor.

Physical Components

Conveyor

- Moves bottles through the labeling station.

Product Detect Sensor

- Detects bottle arrival.
- Starts the labeling sequence.

Print Engine

- Generates or confirms label print completion.

Label Feed Assembly

- Advances label stock into position.

Label Ready Sensor

- Confirms the label is available for application.

Tamp Cylinder

- Extends the tamp mechanism toward the bottle.

Tamp Pad

- Contacts or transfers the label onto the bottle surface.

Label Applied Sensor

- Confirms the label application event.

Tamp Home Sensor

- Confirms the tamp mechanism returned to home position.

Controller

- Coordinates machine sequence and timing.

Expected Interaction Model

Bottle Detect

->

Print Complete

->

Label Ready

->

Tamp Extend

->

Label Applied

->

Tamp Home

Interaction Explanation

During a normal cycle, the conveyor presents a bottle to the product detect sensor. The controller coordinates print completion, label feed readiness, tamp extension, label application confirmation, and tamp return before the next bottle is processed.

Monitored Relationships

- Bottle Detect -> Print Complete
- Print Complete -> Label Ready
- Label Ready -> Tamp Extend
- Tamp Extend -> Label Applied
- Label Applied -> Tamp Home

LineAlert Investigation Package

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Machine
Bottle Labeling Machine

Issue
Label Alignment Off

Observed Evidence

- 2026-06-14T05:16:00.000Z Bottle Detect
- 2026-06-14T05:16:00.072Z Print Complete
- 2026-06-14T05:16:00.118Z Label Ready
- 2026-06-14T05:16:00.240Z Tamp Extend
- 2026-06-14T05:16:00.393Z Label Applied
- 2026-06-14T05:16:00.520Z Tamp Home
- Tamp Extend observed at 240 ms
- Label Applied observed at 393 ms
- Observed lag calculated from deterministic cycle data

Expected State
Tamp Extend -> Label Applied within 100 ms

Observed State
Tamp Extend -> Label Applied in 153 ms

Relationship Analysis

- Expected relationship: Tamp Extend precedes Label Applied by 100 ms
- Observed relationship: Tamp Extend precedes Label Applied by 153 ms
- Relationship integrity: Sequence valid; timing relationship shows moderate drift

Timing Analysis

- Expected lag: 100 ms
- Observed lag: 153 ms
- Delta: +53 ms

Drift Analysis

- No Drift: 0-10 ms delta
- Minor Drift: 11-30 ms delta
- Moderate Drift: 31-75 ms delta (observed)
- Severe Drift: >75 ms delta

Candidate Hypotheses

1. Label guide loosened - highest ranked possibility; could shift label path before tamp contact.
2. Tamp pad wear - could reduce consistent label transfer and add application delay.
3. Product positioning variance - could change bottle presentation during tamp extension.
4. Sensor contamination - could delay readiness or applied-state confirmation.

Peter Troubleshooting Guide

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Guide: Peter Label Alignment Guide

Issue: Label Alignment Off

Prioritized Workflow

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1. Inspect Label Guide - Check the label guide for looseness or misalignment.
2. Inspect Tamp Pad - Check tamp pad condition and label transfer consistency.
3. Verify Product Stop Position - Check bottle presentation and stop repeatability.
4. Inspect Sensors - Check readiness and applied-state sensor transitions.

Troubleshooting Workflow

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Guide:
Peter Label Alignment Guide

Current Recommended Starting Step:
Inspect Label Guide

Reason:
Highest ranked hypothesis:
Label guide loosened

Supporting Evidence:

- Expected lag 100 ms
- Observed lag 153 ms
- Drift +53 ms

Verification Steps

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Verification Required:

Run 10 bottles and confirm alignment within tolerance
Confirm alignment restored

Step Details:

- Step 1: Inspect Label Guide
 - Pass: Continue to tamp pad inspection if alignment drift persists.
 - Fail: Secure and realign the label guide before rebaseline.
 - Verify: Run 10 bottles and confirm alignment within tolerance
- Step 2: Inspect Tamp Pad
 - Pass: Continue to product stop position verification.
 - Fail: Clean or replace tamp pad before rebaseline.
 - Verify: Observe consistent label transfer
- Step 3: Verify Product Stop Position
 - Pass: Continue to sensor inspection.
 - Fail: Stabilize bottle stop position before rebaseline.
 - Verify: Measure bottle stop repeatability
- Step 4: Inspect Sensors
 - Pass: Document results and monitor next production run.
 - Fail: Clean and validate sensors before rebaseline.
 - Verify: Sensor transitions occur within expected timing window

Recommended Actions

1. Inspect and secure the label guide before changing timing parameters.
2. Check tamp pad face condition and replace if wear or residue is observed.
3. Verify bottle stop, spacing, and conveyor presentation at the applicator station.
4. Clean and validate label-ready and applied-state sensors before rebaseline.

Escalation Guidance

Escalate to Operator Lead if moderate drift repeats after guide, tamp, and sensor checks.

Confidence Assessment

Medium-high: deterministic timing data shows a valid sequence with +53 ms drift; root cause is not confirmed.

Tradeoff Summary

Mechanical and sensor checks are low-risk first steps; changing timing baselines before inspection could mask the underlying cause.